

2026 IEEE SUSTAINABLE POWER AND ENERGY CONFERENCE

iSPEC 2026

December 4–6, 2026 | Mar Baselios College of Engineering and Technology, Thiruvananthapuram, Kerala, India



SPECIAL SESSION 3 (SS3) • AI-SPEM

AI-Driven Sustainable Power Systems and Electric Mobility

— Call for Papers —

Organized and Co-Chaired by:

- **Dr. Muthukumari S.** — Postdoctoral Research Associate, Amrita School of Artificial Intelligence, Amrita Vishwa Vidyapeetham, Coimbatore
- **Dr. Nitha P. K.** — Assistant Professor, Amrita School of Artificial Intelligence, Amrita Vishwa Vidyapeetham, Coimbatore
- **Dr. Parvathy S.** — Assistant Professor, Amrita School of Engineering, Amrita Vishwa Vidyapeetham, Coimbatore

Session Overview:

Artificial intelligence offers powerful capabilities for addressing the variability, low inertia, stability, protection, and real-time decision-making challenges introduced by renewable integration, distributed energy resources, and electric mobility. This session invites contributions demonstrating how emerging AI paradigms — machine learning, reinforcement learning, explainable and physics-informed AI — can enable smarter operation, accelerate renewable integration, and improve resilience toward sustainable, carbon-neutral power systems.

Topics of Interest Include, but Are Not Limited to:

- AI-driven load, renewable generation, and inertia forecasting
- Reinforcement learning for power system operation, control, and energy management
- AI-based fault detection, fault classification, protection, and predictive maintenance
- Explainable, trustworthy, and physics-informed AI for power system planning and operation
- Generative and agentic AI for power system simulation and intelligent decision support
- Intelligent energy management for microgrids, virtual power plants, and distributed energy resources
- AI for coordinated EV charging, charging-infrastructure planning, and vehicle-to-grid (V2G) integration
- AI-based EV battery health, state-of-charge/state-of-health estimation, degradation, and thermal safety
- AI-driven cybersecurity and resilience enhancement, including false data injection attack detection and mitigation
- AI-enabled self-powered IoT sensing and energy harvesting for grid monitoring
- AI for electricity market analytics and carbon-aware operation

Important Dates:

Paper Submission Opens	25 August 2026
Paper Submission Deadline	30 September 2026
Acceptance Notification	15 October 2026
Registration Due	1 November 2026
Conference Dates	4–6 December 2026

Format: Half-day technical special session, targeting ~8–10 peer-reviewed papers. Author guidelines and peer-review standards will follow the regular iSPEC 2026 paper submission process. Accepted and presented papers will be submitted for inclusion in IEEE Xplore, subject to IEEE Xplore's scope and quality requirements.

Submission Portal: <https://ispec2026.org/submission>

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